

AMD Instinct

ROCm

vLLM

Hugging Face

Qwen

ROCmBench Agent

A deployment optimization agent for open-source LLMs on AMD GPUs. It turns raw benchmark runs into production-ready serving decisions.

BEST 7B BOUNDARY

c24

3,392 tok/s @ 1.667s p95

GPU PROOF

100%

peak ROCm SMI utilization

DECISION

gate

deploy / warn / block

Running a model is easy. Trusting a config is not.

- Tokens/sec alone hides tail latency, error behavior, VRAM pressure, and cost.
- Manual benchmark notebooks are hard to repeat and weak as deployment evidence.
- Teams need a production gate: what should ship, what needs another run, and why.

WRONG METRIC

fastest

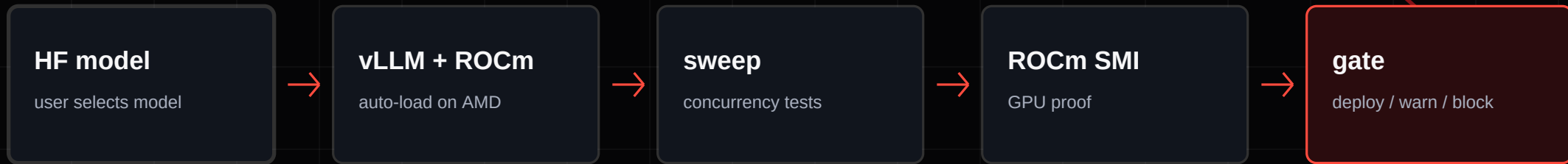
is not always deployable

RIGHT QUESTION

trust?

would this survive prod load

The agent moves from model choice to deployment decision.



The live path auto-loads the selected Hugging Face model on the AMD droplet, waits for `/v1/models` to confirm the exact model, then benchmarks only after the endpoint is valid.

The score is built for production, not leaderboard screenshots.

THROUGHPUT

throughput

tokens/sec + request rate

TAIL LATENCY

tail

avg + p95 latency

RELIABILITY

reliability

error-rate penalty

COST

cost

GPU-hour to \$/1M tokens

ROCM PROOF

ROCm

utilization + VRAM saturation

DECISION

decision

deploy / warn / block

Qwen 7B found the exact point where faster stopped meaning better.

Qwen2.5-7B: throughput vs tail latency

p95 1.695s

p95 1.667s

p95 3.249s

2,690

3,392

3,423

c16

c24

c32

Concurrency 32 adds only 0.9% throughput over c24, but p95 latency nearly doubles.

Agent decision

Pick concurrency 24 for balanced production serving. Concurrency 32 barely improves throughput, but pushes p95 from 1.667s to 3.249s.

The follow-up suite turned one benchmark into real deployment evidence.

TEST	MODEL	FINDING	PROOF
Concurrency ceiling	Qwen 7B	c24 wins balanced	3,392 tok/s
8192 context	Qwen 7B	c16 remains stable	1.523s p95
Soak stability	Qwen 7B	0% errors	2,689 tok/s
Model tradeoff	Qwen 3B	higher throughput, worse p95	4,219 tok/s

What ships in the app

- HF search + model selection with live vLLM auto-load.
- Concurrency sweeps and OpenAI-compatible endpoint benchmarks.
- ROCm SMI monitor parsing and GPU saturation gate.
- Side-by-side model/config comparison across saved and live runs.
- Generated deployment report and vLLM launch command.

OPEN SOURCE

MIT

GitHub-ready project

STACK

AMD

MI300X + ROCm + vLLM

CLOSE

Benchmarking is easy. Deployment decisions are the product.

ROCmBench Agent turns AMD GPU experiments into a repeatable launch gate: load the model, test the configs, prove GPU behavior, and choose what should ship.

PagedAttention scoring

speculative decoding tests

multi-GPU tensor parallel

CI regression gate

github.com/naksshhh/ROCmBench-Agent